

Lecture 18: Poisson Process

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—title: “Lecture 18: Poisson Process”author: “Jamie Haddock”format: revealjs: output-file: Lecture18_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

1	The Poisson process::: {.callout-note icon=false}## Definition: Poisson processThe continuous-time limit of a Bernoulli process, rate λ . $N(t)$ = number of arrivals by time t : $N(t) \sim \text{Poisson}(\lambda t)$, and interarrival times are iid $\text{Exponential}(\lambda)$ (memoryless).::: .	
	FactIndependent/stationary increments: counts over disjoint intervals are independent, and their distribution depends only on interval length. Merging: independent $\text{Poisson}(\lambda_1)$, $\text{Poisson}(\lambda_2)$ processes union to $\text{Poisson}(\lambda_1 + \lambda_2)$. Splitting: routing each arrival independently with prob. q yields independent $\text{Poisson}(\lambda q)$, $\text{Poisson}(\lambda(1 - q))$.:::—## Simulating a Poisson process	

```
lam, T = 3.0, 20.0interarrivals = rng.exponential(1/lam, size=2000)arrival_times = np.cumsum(interarrivals)
from scipy.stats import poissoncounts = [np.sum(rng.exponential(1/lam, size=200).cumsum() < T) for _ in range(1000)]
```

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